

VTU Machine Learning Module 3 - Comprehensive Notes

1. Nearest Neighbour Learning

Definition:

A non-parametric method used for classification and regression. It predicts the label of a new instance based on the majority label of its k nearest neighbors.

Key Concepts:

- Distance Metric: Commonly uses Euclidean distance.
- Advantages: Simple, effective for small datasets.
- Disadvantages: Computationally expensive for large datasets, sensitive to irrelevant features.

Formula:

- Euclidean Distance: $d(x, y) = \sqrt{\sum (x_i - y_i)^2}$

Example Problem:

Q: Given training points A(1,2), B(2,3), C(3,3) with labels 0, 1, 1 respectively. Predict the label for point P(2,2) using 3-NN.

A: Distances: A=1.0, B=1.0, C=1.41 $\rightarrow$ Nearest: A, B, C $\rightarrow$ Labels: 0,1,1 $\rightarrow$ Majority = 1 $\rightarrow$ Predicted label = 1

2. Weighted KNN Algorithm

Definition:

A variation of KNN where closer neighbors have more influence on the prediction.

Key Concepts:

- Weighting Function: Commonly inverse of distance ($w = 1/d$).
- Use Case: Improves accuracy when closer neighbors are more relevant.

Example Problem:

Q: Using the same data as above, apply weighted KNN with weights = $1/\text{distance}$.

A: Weights: A=1.0, B=1.0, C=0.71 $\rightarrow$ Weighted sum: $0*1 + 1*1 + 1*0.71 = 1.71 \rightarrow$ Predicted label = 1

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3. Nearest Centroid Classifier

Definition:

Classifies based on the closest class centroid (mean of feature vectors for each class).

Formula:

Assign x to class c where $c = \operatorname{argmin} \|x - \mu_c\|$, μ_c is the centroid of class c .

Example Problem:

Q: Class 0: (1,2), (2,2); Class 1: (4,4), (5,5). Predict class for (3,3).

A: $\mu_0 = (1.5, 2)$, $\mu_1 = (4.5, 4.5)$; Distances: to $\mu_0 \sim 1.8$, to $\mu_1 \sim 2.12 \rightarrow$ Predicted class = 0

4. Locally Weighted Regression (LWR)

Definition:

A regression method that fits a model locally around the query point.

Key Concepts:

- Assigns weights to training examples based on distance to query point.
- Weight Function: Gaussian kernel $w(x) = \exp(-\|x - x_i\|^2 / (2\tau^2))$

Example Problem:

Q: Given points (1,1), (2,2), (3,3), predict y at $x=2$ using LWR with $\tau=1$.

A: Compute weights using Gaussian kernel, fit weighted linear regression locally around $x=2$.

5. Regression Methods

Linear Regression:

- Hypothesis: $h(x) = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$
- Cost Function: $MSE = (1/n) * \sum (y_i - h(x_i))^2$

Multiple Linear Regression:

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- Extension of linear regression with multiple input features.

Polynomial Regression:

- Adds polynomial terms to model non-linear relationships.
- Example: $y = w_0 + w_1x + w_2x^2 + \dots + w_nx^n$

Logistic Regression:

- Used for binary classification.
- Hypothesis: $h(x) = 1 / (1 + e^{(-z)})$, where $z = w^T * x$
- Cost Function: $- [y \log(h(x)) + (1-y) \log(1-h(x))]$

Example Problem:

Q: Fit a linear regression line for points (1,2), (2,3), (3,5).

A: Use least squares to find best-fit line: $y = ax + b \rightarrow a \sim 1.5, b \sim 0.33$

6. Decision Tree Learning

ID3 Algorithm:

- Uses Information Gain to select best attribute.
- Entropy: $H(S) = -\sum(p_i * \log_2(p_i))$
- Information Gain: $IG(S, A) = H(S) - \sum(|S_v|/|S| * H(S_v))$

C4.5 Algorithm:

- Extension of ID3, handles continuous attributes.
- Uses Gain Ratio = $IG(S, A) / \text{SplitInfo}(A)$

CART Algorithm:

- Uses Gini Index: $\text{Gini}(S) = 1 - \sum(p_i^2)$
- Produces binary trees.

Regression Trees:

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- Predicts continuous values.
- Splits data to minimize variance within subsets.

Example Problem:

Q: Given data with attribute Outlook (Sunny, Overcast, Rain) and Play (Yes/No), compute IG for Outlook.

A: Compute entropy of dataset and conditional entropies for each Outlook value, then $IG = H(S) - \text{Sum weighted } H(S_v)$